

OmniRNA: A Discovery Challenge for RNA Foundation Models

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1 Problem Description

RNA plays a central role in molecular biology. It participates in every stage of gene regulation, from transcription and processing to export, translation, and degradation. Well-characterized protein complexes mediate these stages. However, it remains unknown why the same complexes produce diverse RNA variants with differing splicing isoforms, export efficiencies, and abundances. This mystery leads to a key hypothesis: RNA molecules contain distinct intrinsic codes—embedded within their sequence and structure—that actively determine their biological fates.

Foundation models (FMs) offer a promising route to decode these signals. They learn from large amounts of unlabelled biological data and can be adapted to downstream tasks with limited supervision. This paradigm has already delivered strong results in genomics (e.g., Evo2 [1] and AlphaGenome [2]) and protein modelling (e.g., ESM3 [3]). However, despite emerging progress [4, 5], RNA remains fundamentally harder to model [6]. Unlike DNA, RNA molecules are not static lines of code. They fold into complex secondary (2D) and tertiary (3D) structures *in vivo*. These structures are not merely passive geometries; they actively dictate biological traits. Furthermore, RNA regulatory mechanisms evolved early and are deeply conserved. RNA annotations are also scarcer and less standardised than those for DNA and proteins. As a result, strong genomic and protein models cannot simply be repurposed to decipher the RNA functional code.

Current RNA benchmarks do not fully answer this question. Most focus on a single narrow task, such as sequence classification or structure prediction. This makes it difficult to assess whether a model has learned a general and biologically meaningful representation of RNA. In practice, a strong RNA model should do three things well. It should predict the effects of mutations, recover RNA structure, and capture evolutionary relationships.

OmniRNA addresses this gap. This discovery challenge is organised around three pillars of RNA biology: mutational fitness, RNA structure, and evolution. We challenge the community to build models that can (1) predict the effects of genetic variants on RNA fitness, (2) infer both secondary and tertiary structure from sequences, and (3) identify evolutionary signals that preserve function across divergence. These capabilities matter for RNA therapeutics, synthetic biology, variant interpretation, and comparative genomics. The challenge will provide a standard and reproducible evaluation setting for AI models in RNA biology.

2 Tasks and data

OmniRNA contains three tasks: RNA fitness prediction, RNA structure prediction, and RNA evolutionary reasoning. For each task, we will release training, validation, and development-test splits. A separate hidden test split will be used for the final ranking. All data, documentation, starter code, and evaluation scripts will be hosted on Codabench¹ and mirrored through a public GitHub repository.

2.1 RNA Fitness Prediction

This task measures whether a model can predict the functional effect of RNA variation. The data are collected from public RNA mutational scanning studies. Each example contains a sequence variant and

¹<https://www.codabench.org/>

an experimentally measured fitness value. We will include both in-distribution and out-of-distribution splits. In-distribution splits use random partitioning within a study. Out-of-distribution splits hold out higher-order mutants or extreme-fitness variants. This design tests whether models can extrapolate beyond the local mutational neighbourhood of the training data. Metadata will include RNA type, mutation count, study identifier, and sequence length when available.

2.2 RNA Structure Prediction

This task measures whether a model can recover RNA secondary and tertiary structures from sequence. The benchmark is built from the Protein Data Bank [7]. We first extract all entries containing RNA and separate each RNA chain, resulting in 24,512 chains with corresponding 3D structures. We keep structures with resolution better than 4Å and sequence length between 20 and 500 nucleotides. We then collect newly released RNA structures published after 1 January 2026. This produces 389 candidate structures. To avoid leakage, we remove any structure with more than 80% sequence similarity to the training set. This leaves 146 non-redundant tertiary structures, from which we derive the corresponding 2D structures using DSSR [8]. We split these into a public development-test set and a hidden final-test set. This protocol tests generalisation to newly determined RNA structures rather than memorisation of known folds. Metadata will include sequence length, resolution, experimental method, and RNA type when available.

2.3 RNA Evolutionary Reasoning

This task measures whether a model captures evolutionary constraints that preserve RNA function despite sequence divergence. We use Rfam families [9] to create two subtasks. The first subtask is quartet phylogeny classification. Given four RNA sequences from distinct taxa, the model must choose the correct unrooted tree topology from three candidates. The second subtask is covariation detection. Here the model must detect compensatory mutation patterns that preserve structural pairing. We construct family-disjoint training, validation, development-test, and hidden-test splits. Families that appear in testing are fully excluded from training and validation. This prevents homologous leakage and tests generalisation to unseen RNA lineages. Metadata will include family identifier, taxonomic group, sequence length, and alignment depth when available.

3 Evaluation

Each task will have a public development leaderboard and a hidden final leaderboard. The public leaderboard will support iteration during the competition. The hidden leaderboard will determine the official ranking. We will also maintain per-task leaderboards and one overall leaderboard.

- For RNA fitness prediction, the public leaderboard will report Spearman correlation and R^2 . The official task score will be the unweighted mean Spearman correlation across all fitness datasets and evaluation splits. We choose Spearman as the primary metric because participants often need to rank variants rather than estimate absolute assay values. R^2 will be reported as a secondary diagnostic metric.
- For RNA tertiary structure prediction, the public leaderboard will report TM-score and RMSD. The official task score will be the mean TM-score across evaluation molecules. We choose TM-score as the primary metric because it is length-normalised and more stable than RMSD for global structural comparison. RMSD will be reported as a secondary metric. For RNA secondary structure prediction, we primarily use the F1 score to evaluate base-pair prediction accuracy, as it provides a balanced measure of precision and recall.
- For RNA evolutionary reasoning, quartet phylogeny classification will be scored with accuracy. Covariation detection will be scored with macro-F1. The official evolution score will be the arithmetic mean of these two subtask scores. This prevents strong performance on one subtask from masking failure on the other.

The overall challenge score will be the arithmetic mean of the three primary task scores after mapping Spearman correlation to the $[0, 1]$ interval as $(1 + \rho)/2$. TM-score, accuracy, and macro-F1 already lie in $[0, 1]$. This produces one transparent final ranking while keeping the three tasks equally important.

To ensure fairness, public leaderboards will use only development-test splits. Final rankings will use hidden test data. Submission frequency will be limited to reduce leaderboard overfitting. Participants

may use public pretrained models and public unlabeled RNA sequence resources, but they must disclose all external resources. Participants may not use external labelled data that overlap with the released benchmark, and they may not use any information derived from hidden test sets or from RNA tertiary structures released after the test freeze date. Final evaluation will require executable Docker submissions.

4 Communication Plan

We will promote the challenge through both machine learning and RNA biology communities. We will use the ECML/PKDD communication channels, relevant mailing lists, social media, lab networks, and research communities in bioinformatics, computational biology, and foundation model evaluation. We will also contact groups that work on RNA language models, RNA structure prediction, and comparative genomics.

To lower the barrier to entry, we will release clear task descriptions, a starter kit, baseline code, example submissions, and a public FAQ. We will maintain a discussion forum on Codabench and a linked GitHub issue tracker. We will communicate results through the ECML/PKDD challenge session, the challenge webpage, and a short post-challenge report.

5 Technical Details

The challenge will run on Codabench. The public GitHub repository will host the starter kit, data loaders, baseline implementations, evaluation scripts, and submission templates. We will provide at least two baselines: one lightweight sequence model [10] and one pretrained RNA model baseline [5].

During the competition, participants will submit predictions to the public leaderboard. For the final phase, shortlisted teams will submit Docker containers with inference code and model weights. The evaluation server will run these containers in a controlled environment. This ensures that final results are reproducible and directly comparable across teams.

6 Reporting Requirements

Every final submission must include: (1) runnable Docker code for inference, (2) model weights, (3) a short method sheet, and (4) a statement that lists all external data, pretrained models, and compute resources used. These items are mandatory for eligibility on the final leaderboard. An optional technical report may accompany public leaderboard submissions. A full solution paper will be required only from teams that accept an invitation to the ECML/PKDD workshop proceedings.

The method sheet must describe the model architecture, training recipe, inference procedure, and resource requirements. It must also state whether the submission uses a foundation model, task-specific training, retrieval, or any hybrid method. The challenge is open to any reproducible computational method. It is not restricted to deep learning only.

7 Timeline

As summarized in Table 1, the proposed timeline is aligned with recent ECML/PKDD Discovery Challenge schedules.

Table 1: Proposed timeline of the OmniRNA challenge.

Event / Phase	Date
Competition Start	15 April 2026
Competition End	01 June 2026
Publish Results	15 June 2026
Camera Ready submission	15 July 2026

8 Privacy and Ethics

The benchmark uses public RNA sequences, public structure data, and public family annotations. It does not include personally identifiable information, clinical records, or patient-level data. Privacy

risks are therefore minimal.

We will still document data provenance, licences, and preprocessing decisions for each task. We will release dataset cards and task cards. We will also describe known limitations, such as assay bias, taxonomic imbalance, and the scarcity of high-resolution RNA 3D structures. The challenge does not ask participants to generate harmful biological sequences or optimise pathogenic function. It focuses on predictive modelling on public benchmark data.

9 Awards

We will recognise the top teams on the overall leaderboard and on each task leaderboard. We will invite and review the top two solution papers for submission to the ECML/PKDD workshop proceedings, subject to conference policy. We will also provide digital certificates for the winning teams.

10 Organisers

Ke Li (k.li@exeter.ac.uk) will serve as the lead organiser and contact person for the challenge. His research covers AI for life sciences, and multi-objective optimisation. He will oversee the scientific framing of the challenge, the benchmark design, and the final evaluation process.

Jack Cole, **Yuan Li**, **Wanli Ma**, and **Tianmeng Hu** are members of the organising team and will support the day-to-day delivery of the challenge. Their responsibilities will include dataset preparation, baseline implementation, documentation, Codabench deployment, submission checking, leaderboard maintenance, and participant communication.

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